4. Review of linearization techniques

Recall that the best linear approximation of a function of 1 variable $f(x)$ near a point $x = a$ is

$$f(x) \approx f(a) + f'(a)(x-a). \tag{5.15}$$

Recall that for a function of 2 variables $f(x,y)$, the best linear approximation near a point $(x,y) = (a,b)$ is

$$f(x,y) \approx f(a,b) + \frac{\partial f}{\partial x}(a,b)\;(x-a) + \frac{\partial f}{\partial y}(a,b)\;(y-b) \tag{5.16}$$

In the course *Differential equations: 2x2 systems*, we saw that to linearize a 2×2 system of autonomous equations near a critical point $(x,y) = (a,b)$,

$$\dot{x} = f(x,y) \tag{5.17}$$

$$\dot{y} = g(x,y), \tag{5.18}$$

we need to replace **both** $f(x,y)$ and $g(x,y)$ by the best linear approximations near (a,b), which are given by

$$f(x,y) \approx$$
$$f(a,b) + \left.\frac{\partial f}{\partial x}\right|_{(a,b)} (x-a) + \left.\frac{\partial f}{\partial y}\right|_{(a,b)} (y-b) \;=\; f(a,b) + f_x(a,b)\;(x-a) + f_y(a,b)\;(y-b)$$

$$g(x,y) \approx$$
$$g(a,b) + \left.\frac{\partial g}{\partial x}\right|_{(a,b)} (x-a) + \left.\frac{\partial g}{\partial y}\right|_{(a,b)} (y-b) \;=\; g(a,b) + g_x(a,b)\;(x-a) + g_y(a,b)\;(y-b).$$

Using the definition of matrix multiplication, the best linear approximation of the vector-valued function $\begin{pmatrix} f(x,y) \\ g(x,y) \end{pmatrix}$ near the point $(x,y) = (a,b)$ is

$$\begin{pmatrix} f(x,y) \\ g(x,y) \end{pmatrix} \approx \underbrace{\begin{pmatrix} f(a,b) \\ g(a,b) \end{pmatrix}}_{\text{value at } (a,b)} + \underbrace{\left.\begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}\right|_{\mathbf{a}}}_{\text{derivative at } (a,b)} (\mathbf{x}-\mathbf{a}), \qquad \text{where } \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} a \\ b \end{pmatrix}. \tag{5.19}$$

We can abbreviate this notation as

$$\mathbf{f}(\mathbf{x}) \approx \mathbf{f}(\mathbf{a}) + \mathbf{J}(\mathbf{a})(\mathbf{x}-\mathbf{a}),$$

where

$$\mathbf{f}(\mathbf{x}) = \begin{pmatrix} f(x,y) \\ g(x,y) \end{pmatrix},$$

$$\mathbf{f}(\mathbf{a}) = \begin{pmatrix} f(a,b) \\ g(a,b) \end{pmatrix},$$

$$\mathbf{J} = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix},$$

and the matrix $\mathbf{J}(\mathbf{a})$ is the matrix of partial derivatives evaluated at $\mathbf{a} = \begin{pmatrix} a \\ b \end{pmatrix}$.

The matrix $\mathbf{J}$ is called the **Jacobian matrix** .

To determine the stability of the system at the critical points, you evaluate the Jacobian at each critical point. If both of the eigenvalues of the Jacobian at a point **a** have negative real part, the critical point is stable. If there is an eigenvalue with positive real part, the critical point is unstable. (However, if the highest real part among eigenvalues is exactly 0, one cannot determine the stability just from the linear approximation of the system.)

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